BDA400 • DATA SCIENCE TOOLS AND TECHNIQUES

TECHNICAL ANALYSIS USING R

Assignment 5 | Development Phase



STUDENT	Lubna Afandi
COURSE	BDA400 - Data Science Tools and Techniques
ASSIGNMENT	Technical Analysis Using R - Development Phase (15%)
SYMBOLS	NVDA • AMZN • TSLA
RESULT SNAPSHOT	Observed through 30 October 2026

PROJECT REPOSITORY

Open Lubna’s Assignment 5 folder on GitHub

Nine base-R indicator functions • Validation checks passed • Seven generated result CSVs

01 | Project overview

This report documents the Assignment 5 R project, its indicator implementations, execution workflow, and the results observed in RStudio. The project applies nine technical-analysis functions to daily NVDA, AMZN, and TSLA price data. Indicator calculations use standard R functions; quantmod is used only to retrieve Yahoo Finance data.

What was completed

The project contains one R file per required indicator, a reusable filtering/trading-rules file, an input-preparation script, a main runner, and a validation script. The validation run displayed “All Assignment 5 indicator checks passed.” The main runner completed for all three symbols and exported the result CSVs to the Assignment 5 **outputs/** folder.

Repository structure

PATH	PURPOSE
functions/01_sma.R through 09_crossunder.R	Nine individual indicator implementations.
functions/10_filters_and_trading_rules.R	Recent-observation filter and Long/Flat position logic.
00_prepare_inputs.R	Retrieves daily OHLCV data for NVDA, AMZN, and TSLA; saves input CSVs in outputs/.
99_validate_functions.R	Runs deterministic checks for indicator behavior and output shapes.
00_run_assignment5.R	Loads functions, processes all symbols, and exports analysis, signals, and combined summary CSVs.
README.md and outputs/	Project/run documentation and generated data/results.

The GitHub folder link above points to the Assignment5 project directory. The report reflects the updated local workflow in which both input and generated result CSV files are stored in Assignment5/outputs/.

02 | Indicator implementation guide

Each function is kept in a separate script and follows the supplied assignment naming pattern. The descriptions below summarize the implemented calculation and returned value.

FUNCTION / FILE	CALCULATION SUMMARY	OUTPUT
sma() • 01_sma.R	Arithmetic mean for each complete trailing window; early dates are padded with NA by the runner for date alignment.	Aligned numeric series.
ema() • 02_ema.R	Recursive exponential average using multiplier 2/(period+1), seeded by the first observation.	Numeric series aligned to input.
macd() • 03_macd.R	Difference of short and long EMAs, its signal EMA, and the difference between MACD and signal.	MACD, signal, histogram.
stdev() • 04_stdev.R	Population standard deviation: square root of the mean squared deviation from the mean (divide by n).	One numeric value.
linreg() • 05_linreg.R	Least-squares slope and intercept for the selected source segment and offset.	Slope, intercept, predictions.
rsi() • 06_rsi.R	Relative Strength Index based on gains/losses with Wilder smoothing.	Aligned RSI series.
stoch_rsi() • 07_stoch_rsi.R	Normalizes RSI across a rolling range and smooths the %K and %D lines.	%K and %D series.
crossover() • 08_crossover.R	Detects upward or downward changes in the relative position of two aligned series.	Up, Down, or None.
crossunder() • 09_crossunder.R	Detects when the first series crosses below the second series.	True/False signal series.

Filter and trading rule

For each symbol, the runner selects the latest 90 calendar days and retains observations whose Close is above the independently calculated SMA20. The position rule uses SMA20 and SMA50: an upward crossover changes the position to Long, while a downward crossover returns it to Flat. These are instructional signals generated by the supplied rule, not investment advice.

Calculation conventions

The runner aligns moving-average values with the original dates. RSI uses Wilder smoothing and leaves the initial unavailable values as NA. MACD uses 12-, 26-, and 9-period EMAs in the run shown. Standard deviation follows the assignment’s population formula. Edge cases such as zero gain/loss ranges are handled explicitly by the functions.

03 | Reproducible run instructions

Run the steps from the Assignment5 folder in RStudio. Keep the project files together so the relative paths resolve.

STEP	ACTION	EXPECTED RESULT
1	Set the working directory to Assignment5.	RStudio Console shows the Assignment5 working folder.
2	Run **source("00_prepare_inputs.R")**.	Downloads OHLCV data for NVDA, AMZN, and TSLA; input CSVs are saved in outputs/.
3	Run **source("99_validate_functions.R")**.	Console: “All Assignment 5 indicator checks passed.”
4	Run **source("00_run_assignment5.R")**.	Console summary for all three symbols; result CSVs are written to outputs/.
5	Review the files in **Assignment5/outputs/**.	Seven result CSVs plus the three input data CSVs and README.md.

Commands

```
source("00_prepare_inputs.R")
source("99_validate_functions.R")
source("00_run_assignment5.R")
```

Files created by the runner

SYMBOL	TECHNICAL ANALYSIS	FILTERED SIGNALS
AMZN	AMZN_technical_analysis.csv	AMZN_filtered_signals.csv
NVDA	NVDA_technical_analysis.csv	NVDA_filtered_signals.csv
TSLA	TSLA_technical_analysis.csv	TSLA_filtered_signals.csv

Combined summary file: **assignment5_summary.csv**. This is the seventh result CSV. The source inputs (AMZN_data.csv, NVDA_data.csv, TSLA_data.csv) are separate from these seven generated results.

The CSV outputs are machine-readable project artifacts. This PDF summarizes representative latest values from the completed RStudio run; it does not replace the scripts or the CSV files.

04 | Observed results

The RStudio run processed 961 daily observations per symbol, with the latest date shown as 30 October 2026. The table records the values visible in the completed runner output. Values are rounded for readability; the CSV files retain the exported precision.

TICKER	N	LAST DATE	CLOSE USD	SMA20	RSI14	MACD
NVDA	961	2026-10-30	204.64	221.4595	35.05601	-3.657633
AMZN	961	2026-10-30	240.35	254.3905	38.26281	-2.013436
TSLA	961	2026-10-30	399.81	380.6975	60.86296	9.067347

Latest position and filtered rows

SYMBOL	LATEST POSITION	FILTERED ROWS / LAST 90 DAYS
NVDA	Flat	40
AMZN	Long	27
TSLA	Long	47

Brief interpretation

At the captured run, NVDA and AMZN closed below their SMA20 values and had negative MACD readings; the rule reported Flat for NVDA and Long for AMZN. TSLA closed above its SMA20 with a positive MACD reading and a Long position. These outputs describe the implemented rule at that run date only; they are not predictions and can change when new market data are retrieved.

Generated result files (seven)

1. AMZN_filtered_signals.csv	2. AMZN_technical_analysis.csv
3. assignment5_summary.csv	4. NVDA_filtered_signals.csv
5. NVDA_technical_analysis.csv	6. TSLA_filtered_signals.csv
7. TSLA_technical_analysis.csv	

05 | Submission checklist

The assignment instructions require separate R scripts for the indicators, one PDF or Word document with brief implementation explanations, clear run instructions, a cover page with the student name, assignment title, and repository link, and submission of the work in the repository and through the LMS.

REQUIREMENT	EVIDENCE IN THIS PROJECT
Nine named indicator implementations	Nine separate R files in functions/; function names follow the assignment templates.
Core R indicator calculations	Indicator logic uses base R; quantmod is limited to Yahoo Finance data import.
Validation	99_validate_functions.R completed with all indicator checks passed.
Execution instructions	This report lists the working directory and the three source() commands in order.
Cover and repository location	Cover identifies Lubna Afandi, BDA400 Assignment 5, and links to the public Assignment5 folder.
Repository results	Seven generated result CSVs were uploaded to Assignment5/outputs/; the scripts and README are in the Assignment5 folder.
LMS documentation file	Upload this PDF to the LMS using Afandi_Lubna_CA_BDA400_A05.pdf.

Reproducibility note

Yahoo Finance data can change when the script is run again. The results table documents the captured execution on 30 October 2026. A later run may produce updated prices, indicators, signals, or row counts. The inputs and generated outputs should therefore be interpreted as a dated snapshot.

Repository link

https://github.com/lubnaafandi89-coder/TechnicalAnalysis/tree/main/Assignment5

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